Resolution Intelligence

A trusted decision layer for customer-issue operations

Customer signal » Context » Pattern » Priority » Action

Product management and technical build specification. Portfolio prototype demonstrating a governed data foundation on public CFPB data, built with Snowflake, dbt, and Vercel.

Version	1.1 — reconciled against verified source audit
Last updated	August 15, 2026
Owner	Shem Nyachieo
Status	Phase 0 — product contract. No implementation code written.
Authority	The Markdown documents under docs/ are authoritative. This PDF is a presentation artifact regenerated from them.

What changed in v1.1

An independent audit of the live CFPB source on August 15, 2026 disproved several assumptions in v1.0. The public dataset contains **no company response timestamp**, and the **Consumer disputed?** field was removed from CFPB exports in June 2026. The response-duration metric and the entire dispute policy were removed rather than relabelled. See ADR-004 and docs/09_supported_vs_unsupported_metrics.md.

1. Executive positioning

Resolution Intelligence is a **trusted decision layer for customer-issue operations**. It transforms published complaint records into governed issue context, emerging-pattern signals, explainable priority, and recommended investigation or action — each carrying its supporting evidence and its stated limitations.

The prototype uses the public CFPB Consumer Complaint Database. In a real deployment, an organization would connect its own authorized case-management, CRM, knowledge-base, and interaction data.

Hero statement: **Turn customer signals into the next best action.**

What it is not

- Not complaint-management software.

- Not a complaint-resolution prediction engine.
- Not a production financial-services decision engine.
- Not a credit, underwriting, fraud, eligibility, or regulatory decision system.
- Not a consumer scoring system, and not a company performance ranking.
- Not an integration with CFPB, any financial institution, or Twilio.

Why this is Twilio-adjacent

Twilio frames agent productivity around unifying channels and carrying context across AI and human agents. This project works the layer beneath: what is known about an issue, what pattern it belongs to, what is uncertain, and what should happen next. Communications and agents are only as useful as the context behind them. The prototype is *activation-ready*, not activated.

2. Source reality — verified, not assumed

Retrieved and verified August 15, 2026 against the CFPB field reference, release notes, OpenAPI specification, live search API, and bulk CSV archive.

Property	Verified finding
Records	17,119,590 published complaint records
Coverage	2011-12-01 through 2026-08-15
Schema	Exactly 16 fields. No seventeenth field exists.
Update cadence	Daily (CSV archive). The bulk **JSON** archive lags — do not use.
License	CC0
Ingestion	Bulk CSV archive. The filtered API export caps at ~100,000 rows.

There is no company response timestamp

The dataset publishes **date_received** (when CFPB received the complaint) and **date_sent_to_company** — the date CFPB *routed* the complaint to the company. There is no third date. Nothing records when a company received, opened, responded to, or resolved anything.

For modern web-submitted complaints the two dates are separated by **seconds** (observed: 23s, 25s, 13min). A duration derived from them would measure CFPB routing automation while being labelled company responsiveness. The specified **response_days_calendar** metric was removed, and no replacement duration field is permitted.

Fields that do not exist

Field	Status
Consumer disputed?	Discontinued as a filter Nov 2017; **removed from exports June 2026**. Absent from the field reference, both retrieval surfaces, and the OpenAPI schema.
Consumer consent provided	Removed from exports June 2026.

Field	Status
Company response date	Has never existed in the public dataset.
Satisfaction / sentiment	Does not exist in any form.
Company size or denominator	Not published — no rate or comparison is computable.

3. What the product may and may not measure

This register is binding and test-enforced. No model, export, or interface may publish a metric marked **No**. Full version with evidence: docs/09_supported_vs_unsupported_metrics.md.

Metric	Supported?	Caveat
Complaint volume	**Yes**	A count of published complaints, not of customer issues
Issue trends	**Yes**, qualified	Legacy labels versioned, not merged
Product trends	**Yes**, qualified	Evaluated within category — ~81% is credit reporting
Channel trends	**Yes**, low info	96.3% is Web
Published timely response	**Yes**, as category	Not an interval. 99.38% Yes
Emerging issue detection	**Yes**, qualified	Never evidence of a market incident
Consumer disputed signal	**No**	Field does not exist
Actual response duration	**No**	No response timestamp exists
Resolution duration	**No**	Same
Customer satisfaction	**No**	No such field
Root cause	**No**	Categories are not causal analysis
Customer lifetime value	**No**	No customer identity or account data
Individual customer risk	**No**	Prohibited product boundary
Market-wide complaint rate	**No**	No denominator exists

Signal confidence domain

Every derived signal carries one of four values. This is a **qualitative interpretation status**, not a statistical measure — no numerical confidence intervals are manufactured.

Value	Meaning
HIGH	Directly observed source field
MEDIUM	Deterministic derived signal with documented methodology

Value	Meaning
LIMITED	Affected by coverage, publication lag, or denominator limits
NOT_SUPPORTED	A conclusion the public data cannot defensibly establish

4. Interpretation controls

Three measured properties of the source would produce misleading output if left uncontrolled. Each has a named control.

Publication lag

Complaints are published before the company response is necessarily recorded.

Window (by date_received)	Still 'In progress'
Last 7 days	**88.52%**
30–60 days prior	**33.94%**
~12 months prior	0.00%
Whole database	3.55%

Control: recent_publication_lag_flag — set on records within a seed-configured trailing **60-day** window, or with response status *In progress* at any age. The window is set from measurement: a third of complaints remain unresolved one to two months after receipt. A high count of *In progress* records is evidence of publication timing, **never** of poor company performance.

Category concentration

Credit-reporting categories account for roughly **81%** of all records. In June 2026 the CFPB stated it “cannot rely upon the consumer complaint portal data as a reliable reflection of actual market conditions” absent announced corrections, attributing part of the rise to credit repair organizations misusing the complaint process.

Control: trends evaluate *within* product category, require a minimum baseline volume, and publish **observed_share_pct** alongside percentage change. A large percentage increase is never surfaced as evidence of a market incident.

Observed, not representative

The database is an **observed public complaint dataset** — complaints that met publication criteria. Referred complaints and small depository institutions are structurally absent. It is not a sample of consumer experience, and no output may imply market-wide prevalence, harm, or dissatisfaction.

5. Architecture

Raw public records are immutable; dbt owns transformations; policy is data-driven through seeds; final models are explainable; the application reads a curated read-only export — never raw tables.

Stage	Detail
Ingestion	Official bulk CSV archive » schema validation (exactly 16 columns) » source retrieval record
Storage	Snowflake: RAW, ANALYTICS_DEV, ANALYTICS_PROD, GOVERNANCE
Transformation	dbt: sources » staging » intermediate » marts
Policy	Version-controlled seeds — thresholds never buried in SQL
Delivery	Curated versioned export » Next.js on Vercel

Final marts

Model	Grain	Purpose
fct_complaints	Complaint record	Canonical record with lineage
fct_issue_daily_metrics	Date × approved dimensions	Trusted daily metric layer
agent_case_context	Complaint record	Agent-safe context, signals, confidence
resolution_action_queue	Record × recommendation run	Auditable priority queue
operations_overview_metrics	Date × dashboard dimension	Curated aggregate display

Models revised by the source audit

Model	Change
int_complaint_lifecycle	**Renamed** int_complaint_status_context; all timing derivation removed
stg_cfpb_complaints	Null normalization, string complaint_id, masked-ZIP handling, publication-lag flag
int_issue_trends	Baseline volume, observed share, pattern status, within-category evaluation
int_priority_policy_application	Dispute policy removed; publication-lag policy added; confidence propagation
int_company_issue_patterns	Retained but constrained — LIMITED confidence and a denominator limitation on every row
Any duration model	**Must never be created**

6. Decisioning policy

Prioritize service cases and issue patterns, not people. Every recommendation carries a policy ID, action, priority, at least one reason code, its evidence fields, a confidence value, and a limitation where one applies. **“No action” remains a valid outcome.**

Policy	Trigger	Priority	Action
POLICY_UNTIMELY_RESPONSE	Published timeliness = No	HIGH	ESCALATE_REVIEW

Policy	Trigger	Priority	Action
POLICY_EMERGING_ISSUE	Threshold met + all qualification conditions	HIGH	INVESTIGATE_PATTERN
POLICY_PUBLICATION_LAG	Record flagged for publication lag	MEDIUM	REQUIRE_HUMAN_REVIEW
POLICY_INCOMPLETE_CONTEXT	Decision-critical field missing	MEDIUM	REQUIRE_HUMAN_REVIEW
POLICY_STABLE_PATTERN	No trigger active, data sufficient	LOW	STANDARD_HANDLING
POLICY_CRITICAL_COMBINATION	Two or more qualified HIGH triggers converge	CRITICAL	ESCALATE_REVIEW

Escalation to CRITICAL

Removing the dispute policy removed the only single rule producing CRITICAL. Rather than retire the level or attach it arbitrarily, CRITICAL is now an explicit **combination** outcome — the more defensible design. All four conditions must hold:

- Two or more distinct HIGH policies triggered on the same record.
- data_completeness_status = COMPLETE.
- recent_publication_lag_flag = FALSE — publication lag can never escalate to CRITICAL.
- The contributing pattern signal is a QUALIFIED_SIGNAL.

The intersection is expected to be **rare**, which is the intended behavior. A CRITICAL volume that is not rare indicates a configuration error, not a surge. Five tests enforce this.

Removed in v1.1

POLICY_DISPUTED_RESPONSE — its trigger field does not exist in the source. No proxy or substitute dispute signal may be constructed. **POLICY_LATE_RESPONSE** was renamed **POLICY_UNTIMELY_RESPONSE**, and its reason code renamed **PUBLISHED_UNTIMELY_RESPONSE**, so neither can be read as a measured delay.

7. Governance and required disclosures

- “Portfolio prototype built with publicly available CFPB Consumer Complaint Database data.”
- “The prototype does not identify consumers, contact consumers, make financial decisions, or determine complaint outcomes.”
- “Complaint counts are not representative of all consumer experiences and must be interpreted with relevant context, including company size and market share.”
- “Complaint narratives are excluded.”
- “Independent project; not affiliated with or endorsed by CFPB, any financial institution, or Twilio.”

Deliberate exclusions

Excluded	Reason
Complaint narratives	Opt-in, unverified, revocable consent — a snapshot is not reproducible. No NLP, sentiment, or LLM processing.
tags (Older American, Servicemember)	Protected/vulnerable-population attribute. Governance decision, not oversight.
zip_code	Re-identification surface with no MVP requirement; ~5.7% are masks.
Raw source files in Git	1.4 GB and freely re-downloadable. Committing them would defeat the provenance model.

Type and null handling

- **complaint_id is a string**, not an integer — the OpenAPI spec says int64, the live API returns a string.
- **Masked ZIP values containing X** are privacy masks, never real postal codes. Never cast, parse, or geocode them.
- **Literal "None"** in CSV exports must become null — never a legitimate taxonomy category.
- **Unknown values stay unknown.** Never default to a favorable or unfavorable state.

8. Build sequence

Milestone	Deliverable	Human verification gate
1. Charter	Product contract, provenance, dictionary, policy, architecture, limitations, metric register, ADRs	Confirm scope, disclosures, and that no unsupported metric survives
2. Data foundation	Snowflake bootstrap, tested raw load, schema validation	Validate the first download, field names, and retrieval record
3. dbt core	Staging, fct_complaints, taxonomy, tests, docs	Review grain and source-field mapping before build
4. Intelligence	Trend and policy models, agent context, action queue, seeds	Spot-check aggregates against independent SQL
5. Product surface	Curated export, Vercel site, methodology and limitation pages	Review every exposed column against the register
6. Proof	CI, dbt docs, screenshots, demo script, case study	Rehearse the explanation of limitations and tradeoffs

Success measures

Area	Portfolio measure
Data trust	100% of final actions carry policy ID, reason code, and confidence
Usability	A reviewer understands source, logic, limitations, and action in under five minutes
Product quality	No prohibited claims; no direct PII; transparent caveats
Operational signal	Emerging patterns visible with documented thresholds and qualification

Area	Portfolio measure
Technical rigor	Reproducible load » dbt build » demo export path

9. Final positioning

Product	Resolution Intelligence
Category	Customer-issue intelligence and decisioning layer
Core flow	Customer signal » Context » Pattern » Priority » Action
Tagline	Turn customer signals into the next best action.

> *"I built Resolution Intelligence, a Snowflake and dbt decision layer that converts public CFPB complaint records into governed issue context, emerging-pattern signals, and explainable operational next-best actions. It is intentionally not a complaint-management system or a financial decision engine. It demonstrates the trusted data foundation AI and human agents need to resolve customer issues with more context and consistency."*

The constraints are the deliverable

This project's premise is that a data product is only as good as the claims it refuses to make. An independent source audit disproved several of its own founding assumptions, and the response was to remove capabilities rather than soften language — a response-time metric that would have measured a government API's routing speed, and a policy built on a field that no longer exists. What remains is defensible end to end.

Source links

CFPB Consumer Complaint Database	consumerfinance.gov/data-research/consumer-complaints/
Field reference	cfpb.github.io/api/ccdb/fields.html
Release notes	cfpb.github.io/api/ccdb/release-notes.html
Bulk CSV archive	files.consumerfinance.gov/ccdb/complaints.csv.zip
Twilio agent productivity	twilio.com/en-us/solutions/agent-productivity

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